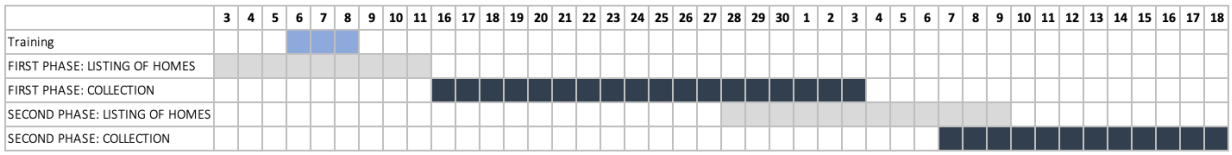
NUMMA

rCARI VALIDATION STUDY

FINAL REPORT

SUMMARY OF RESULTS:

The collection of information, was made during two phases, first phase starting on the 16th of June, and the 2nd phase starting on July 7th. Below, a brief schedule of the different stages made during the study:



The following tables detail the final results for remote and face-to-face collections during the first and second phase of the study. It is important to take into account that these only consider valid results (no duplicates) downloaded from the MoDa platform until July 18.

METHOD	EFFECTIVE	NOT EFECTIVE (REGISTERED IN MODA PLATFORM)	NOT EFECTIVE (NOT REGISTERED IN MODA PLATFORM)	TOTAL LISTING OF HOMES	% COMPLETED
REMOTE	**658**	472	69	1.199	55%
FACE-TO-FACE	**762**	56	355	1.173	65%

					FACE TO FACE			REMOTE		
REGION	**PROVINCE**	**CITY**	**VILLAGE / COMMUNITY**	**TYPE OF VILLAGE**	**ORIGINAL SAMPLE**	**EFFECTIVE**	**%**	**ORIGINAL SAMPLE**	**EFFECTIVE**	**%**
Costa	Guayas	Salitre	EL SALITRE	INCLUDED	28	42	150%	26	32	123%
Costa	Guayas	Salitre	JUNQUILLAL	INCLUDED	32	45	141%	32	39	122%
Costa	Guayas	Salitre	GENERAL VERNAZA	INCLUDED	38	37	97%	40	34	85%
Costa	Santa Elena	Santa Elena	COLONCHE	ELIMINATED	28	29	104%	28	11	39%
Costa	Santa Elena	Santa Elena	MANGLARALTO	INCLUDED	34	54	159%	32	49	153%
Costa	Santa Elena	Santa Elena	SIMÓN BOLIVAR	INCLUDED	34	39	115%	32	44	138%
Sierrra	Imbabura	Cotacachi	SEIS DE JULIO DE CUELLAJE	ELIMINATED	28	19	68%	30	21	70%
Sierrra	Imbabura	Cotacachi	PEÑARRERA	ELIMINATED	23	14	61%	22	14	64%
Sierrra	Imbabura	Cotacachi	APUELA	ELIMINATED	28	16	57%	28	26	93%
Sierrra	Imbabura	Otavalo	SAN JOSE DE QUICHINCHE	ADDED	0	20		0	16	
Sierrra	Imbabura	Otavalo	SAN JUAN DE ILUMAN	ADDED	0	23		0	43	
Sierrra	Imbabura	Cotacachi	IMANTAG	INCLUDED	32	31	97%	35	35	100%
Sierrra	Imbabura	Cotacachi	QUIROGA	INCLUDED	28	52	186%	26	41	158%
Sierrra	Chimborazo	Guamote	CEBADAS	INCLUDED	28	43	154%	28	31	111%
Sierrra	Chimborazo	Guamote	GUAMOTE	ELIMINATED	28	5	18%	31	6	19%
Sierrra	Chimborazo	Colta	COLUMBE	ADDED		27			19	
Sierrra	Chimborazo	Guamote	PALMIRA	INCLUDED	30	49	163%	30	36	120%

Amazonia	Pastaza	Arajuno	ARAJUNO	INCLUDED	42	57	136%	42	58	138%
Amazonia	Orellana	Loreto	LORETO	INCLUDED	54	53	98%	50	41	82%
Amazonia	Orellana	Loreto	SAN VICENTE HUATICOCHA	ELIMINATED	22	23	105%	25	13	52%
Amazonia	Orellana	Loreto	SAN JOSE DE DAHUANO	ELIMINATED	20	17	85%	20	14	70%
Amazonia	Orellana	Loreto	PUERTO MURIALDO	INCLUDED	28	54	193%	28	33	118%
Amazonia	Orellana	Loreto	SAN JOSÉ DE PAYAMINO	ELIMINATED	15	13	87%	15	2	13%
					600	762	127%	600	658	110%
						162			58	

** TYPE OF VILLAGE: INCLUDED (IN PHASE 1), ELIMINATED (FROM PHASE 1), ADDED (to replace villages eliminated)

Below, a summary of the final proportion for each region, for remote and face-to-face:

	REGION	EFFECTIVE REGISTERS	WEIGHT
FACE TO FACE	COSTA	246	32%
	SIERRA	299	39%
	AMAZONÍA	217	28%
		762	100%

	REGION	EFFECTIVE REGISTERS	WEIGHT
REMOTE	COSTA	209	32%
	SIERRA	288	44%
	AMAZONÍA	161	24%
		658	100%

REMOTE DATA COLLECTION

Briefly, we explain our methodology for remote collection:
Each enumerator had a list of homes which have to be contacted in three different ways:

- Regular call
- WhatsApp text
- WhatsApp call

This order changes throughout the day, depending on the village (for example, if we notice a community had more WhatsApp coverage than normal call) and the time frame preferred for the call.

- The calls were made from 7:30am until 7:30pm in order to increase the availability of the respondents.
- The enumerators registered the result of every call in a printed database used for their internal control (call again, voicemail, restricted, out of service, number of attempts, etc.)
- The databases rotated among the enumerators to try to maximize the effective calls. This is a mechanism used to try to adapt to the preferences of the households, depending if they prefer a different interview method, contact form, or gender of the interviewer.
- At least two calls were made per day for each home, in the communities that apply according to the schedule, and depending on the number of registers for every community. One of these calls was made during the time

frame specified by the person listed. This process was repeated for the 3-4 days destinated to each community. **It is important to take into account that these minimum number of calls can be higher since the enumerators generally try a normal call and then a WhatsApp call.

- In order to increase the effective responses, we sent a WhatsApp text to the different homes, introducing ourselves, explaining the context of the study and trying to scheduling the call.
- The face-to-face team helped the remote by "alerting" or scheduling a call.
- For every call made, the enumerator was instructed to register every call to the MoDa platform. Nevertheless, due to their own methodology in trying to increase effectiveness, an important number of calls were not being registered. Therefore, we used our internal platform to monitor the number of calls made. **Below we detail the real number of calls registered.
- Once the respondent agreed to continue with the interview, the enumerator started recording the call and, when finished, the results were immediately sent to the platform MoDa.
- At the end of the day, every enumerator registered their results in our platform: number of calls and indicate whether the call was successful or not.

Using our internal control of number of calls (through our platform), we send an updated summary of the number of calls for effective and not effective registers:

		NUMBER OF REGISTERS	NUMBER OF CALLS
EFFECTIVE		**658**	**2.208**
NOT EFECTIVE	Agreed to be interviewed but later	5	35
	Agreed to be interviewed but no	1	3
	Didn't pick up / no answer	194	1.827
	Refused to be interviewed	72	251
	Other	234	1.587
	No call	35	0
	Total	541	3.703
TOTAL		**1.199**	**5.911**

We did 5.911 calls in total, 2.208 calls for the effective and 3.692 calls for the not effective registers.

The following table shows a summary of the number of effective and not effective registers categorized by the number of calls done:

NUMBER OF CALLS	EFFECTIVE	NOT EFFECTIVE					
		Agreed to be interviewed but later	Agreed to be interviewed but no	Didn't pick up / no answer	Refused to be interviewed	Other	TOTAL
0							35
1	235				13		248
2	120			1	24	1	146
3	88	1	1	22	7	32	151
4	45			28	7	28	108
5	57	1		31	5	40	134
6	46			11	8	39	104
7	10			3		21	34
8	10			6	6	12	34

9	10	3		4	1	20	38
10	11			5		7	23
11	7			6	1	8	22
12	3			14		8	25
13	3			11		6	20
14	2			7		3	12
15	4			12		3	19
16				5		3	8
17	1			5		2	8
18				6			6
19	1			7		1	9
20	3			5			8
21	2			2			4
22				1			1
23				2			2
Total	**658**	**5**	**1**	**194**	**72**	**234**	**1.199**

In this table, we can see that most of the effective registers were completed in the first 3 calls. The not effective registers for homes with less than 5 calls are focused on communities (from the first phase) where there was no phone service at all or for homes with incorrect contact information, or on homes that were called during phase 2 until the sample was completed.

Below, the same analysis but with number of calls:

NUMBER OF CALLS	EFFECTIVE	NOT EFFECTIVE					
		Agreed to be interviewed but later	Agreed to be interviewed but no	Didn't pick up / no answer	Refused to be interviewed	Other	TOTAL
1	235				13		248
2	240			2	48	2	292
3	264	3	3	66	21	96	453
4	180			112	28	112	432
5	285	5		155	25	200	670
6	276			66	48	234	624
7	70	27		21		147	238
8	80			48	48	96	272
9	90			36	9	180	342
10	110			50		70	230
11	77			66	11	88	242
12	36			168		96	300
13	39			143		78	260
14	28			98		42	168
15	60			180		45	285
16				80		48	128

17	17			85		34	136
18				108			108
19	19			133		19	171
20	60			100			160
21	42			42			84
22				22			22
23				46			46
Total	**2.208**	**35**	**3**	**1.827**	**251**	**1587**	**5.911**

Additionally, we show below the number of calls for every community and a short explanation of the methodology used and the complications faced during the collection:

REGION	PROVINCE	VILLAGE	NUMBER OF CALLS	EFFECTIVE REGISTERS	NOT EFFECTIVE REGISTERS	COMMENTS
AMAZONIA	**ARAJUNO**	ARAJUNO	307	58	17	Community with the highest effectiveness.
	LORETO	LORETO	351	41	38	Some parts of Loreto do not have cellular coverage. Also, the community was untrusting to give out information because of previous robberies and extorsions that happened in the community.
		PUERTO MURIALDO	396	33	34	Several calls sent to voicemail; we kept on trying.
		SAN JOSE DE DAHUANO	187	14	10	Political Lieutenant did not allow the face-to-face team to continue with the data collection process and asked for a permit from the provincial delegation. This complicated the collection for remote, but we kept on trying.
		SAN JOSÉ DE PAYAMINO	184	2	16	There is not enough cellular signal coverage with any operator. Households give their contacts for emergencies only.
		SAN VICENTE HUATICOCHA	257	13	14	There is bad signal coverage in some sectors confirmed by face-to-face team.
COSTA	**SALITRE**	EL SALITRE	257	32	22	Users had restricted numbers or calls went directly to voicemail. Face-to-face team helped schedule calls to increase effective registers.
		GENERAL VERNAZA	269	34	20	Households kept changing the time of the scheduled calls and did not answer. Face-to-face team helped schedule calls to increase effective registers.
		JUNQUILLAL	282	39	20	Face-to-face team helped schedule calls to increase effective registers.
	SANTA ELENA	COLONCHE	147	11	21	The Political Lieutenant gave out instructions to the community indicating that they should not provide any information, even by phone. The team was able to complete some registers prior to the inconvenient.
		MANGLARALTO	344	49	24	Low responsiveness from homes. Face-to-face team helped schedule calls to increase effective registers.
		SIMÓN BOLIVAR	387	44	50	Strong resistance to cooperate, homes hung up the phone, asked to change the date and time of the call.
SIERRA	**COLTA**	COLUMBE	99	19	10	
	OTAVALO	SAN JOSE DE QUICHINCHE	112	16	22	
		SAN JUAN DE ILUMAN	106	43	5	
	COTACACHI	APUELA	214	26	10	In certain parts of the community, there is not enough cellular coverage.
		IMANTAG	242	35	41	Face-to-face team helped schedule calls to increase effective registers.
		PEÑAHERRERA	144	14	13	The community only has signal coverage from CNT, which made it difficult to contact the households.
		QUIROGA	553	41	55	Strong resistance to cooperate, homes avoided answering the interview or repeatedly changed the

						date and time of the call. Face-to-face team helped schedule calls to increase effective registers.
		SEIS DE JULIO DE CUELLAJE	175	21	20	The community only has signal coverage from CNT, which made it difficult to contact the households.
	GUAMOTE	CEBADAS	251	31	20	Strong resistance to cooperate, homes avoided answering the interview or repeatedly changed the date and time of the call. Face-to-face team helped schedule calls to increase effective registers.
		GUAMOTE	342	6	34	People from the community go to Riobamba from Wednesday to Sunday to work, making it difficult to contact the homes. On Monday and Tuesday, responsiveness was low.
		PALMIRA	305	36	25	Face-to-face team helped schedule calls to increase effective registers.
TOTAL			**5.911**	**658**	**541**	

FACE TO FACE DATA COLLECTION

The specific methodology used for face-to-face collection involves:

- A constant communication from the Coast, Sierra and Oriente teams and the support from our different supervisors both for face-to-face and remote.
- A digital internal register of the face-to-face homes in order to help the enumerators identify the different homes.
- A guide map for every village in order to optimize the route for every enumerator.
- Apart from the supervisors, our face-to-face team is accompanied by guide supervisors, that were part of the first stage of listing homes, in order to help enumerators identify homes more efficiently.
- A team of enumerators focused on helping the ones that are in the different territories to schedule visits for the next hours or day, in order to make their daily schedule more effective.

The following table shows the results for face-to-face collection:

	NUMBER OF REGISTERS
EFFECTIVE	762
NOT EFFECTIVE	56
NOT REGISTERED	355
Total	1.173

Below, a summary of the listed homes that are not registered in the platform due to complications during the collection (comments specified if relevant) or because the sample was completed.

VILLAGE	**UNREGISTERED HOMES**	**COMMENTS**
QUIROGA	45	Due to parties, the users refused to answer the interview and were also difficult to contact. For the second phase, the team went to the outskirts of the community, due to the blockade from the authorities in the most populated part of the community.
IMANTAG	45	Authorities seemed open to the collection of information at first and asked to participate in a social meeting with local authorities. After this meeting, with no concrete answer, the team continued with the collection for face-to-face and remote, until the authorities blocked completely our presence.
SIMÓN BOLIVAR	43	

GUAMOTE	34	Distrust. Additionally, users were difficult to access because they go to fairs in Riobamba for most days of the week. We propose to collect data in the Columbe community, located in Colta.
LORETO	26	Authorities asked for a permit from the provincial delegation, delaying the data collection and generations distrust in some households. For the second phase, the team went to the outskirts of the community, due to the blockade from the authorities in the most populated part of the community.
SEIS DE JULIO DE CUELLAJE	24	
APUELA	21	
MANGLARALTO	17	
SAN JUAN DE ILUMAN	16	
GENERAL VERNAZA	15	
ARAJUNO	14	
PEÑARRERA	13	
PUERTO MURIALDO	13	
EL SALITRE	13	
PALMIRA	12	
SAN JOSE DE QUICHINCHE	12	
CEBADAS	11	Distrust, households refused to answer the interview.
JUNQUILLAL	10	
SAN JOSE DE DAHUANO	8	Political Lieutenant blocked the data collection process.
COLONCHE	7	The Political Lieutenant blocked the data collection process, generating distrust and lack of willingness among the households.
SAN JOSÉ DE PAYAMINO	5	
SAN VICENTE HUATICOCHA	4	
COLUMBE	3	
Total	**411**	

CLARIFICATIONS

Below, a brief summary of the collection for each community:

REGION	PROVINCE	CITY	VILLAGE / COMMUNITY	OBSERVATIONS / DIFFICULTIES
Costa	Guayas	Salitre	EL SALITRE	No major complications with the collection.
Costa	Guayas	Salitre	JUNQUILLAL	No major complications with the collection.
Costa	Guayas	Salitre	GENERAL VERNAZA	No major complications with the collection.
Costa	Santa Elena	Santa Elena	SIMÓN BOLIVAR	Blockade from local authorities while in second phase of the collection.
Costa	Santa Elena	Santa Elena	COLONCHE	The Political authority blocked the data collection process, generating distrust and lack of willingness among the households.
Costa	Santa Elena	Santa Elena	MANGLARALTO	No major complications with the collection.
Sierrra	Imbabura	Cotacachi	QUIROGA	Due to parties, the users refused to answer the interview and were also difficult to contact. For the second phase, the team went to the outskirts of the community, due to the blockade from the authorities in the most populated part of the community.

Sierrra	Imbabura	Cotacachi	IMANTAG	Authorities seemed open to the collection of information at first and asked to participate in a social meeting with local authorities. After this meeting, with no concrete answer, the team continued with the collection for face-to-face and remote, until the authorities blocked completely our presence.
Sierrra	Imbabura	Cotacachi	SEIS DE JULIO DE CUELLAJE	Community eliminated for the second phase of listing due to lack of cellular coverage which limited the remote collection.
Sierrra	Imbabura	Cotacachi	APUELA	Community eliminated for the second phase of listing due to lack of cellular coverage which limited the remote collection.
Sierrra	Imbabura	Otavalo	SAN JUAN DE ILUMAN	Community added for second phase.
Sierrra	Imbabura	Cotacachi	PEÑARRERA	Community eliminated for the second phase of listing due to lack of cellular coverage which limited the remote collection.
Sierrra	Imbabura	Otavalo	SAN JOSE DE QUICHINCHE	Community added for second phase.
Sierrra	Chimborazo	Guamote	GUAMOTE	Intense distrust from the community. Additionally, users were difficult to access because they go to fairs in Riobamba for most days of the week. We propose to collect data in the Columbe community, located in Colta.
Sierrra	Chimborazo	Guamote	CEBADAS	Distrust, households refused to answer the interview. For the second phase, the team went to the outskirts of the community, due to the blockade from the authorities in the most populated part of the community.
Sierrra	Chimborazo	Colta	COLUMBE	Community added for second phase.
Sierrra	Chimborazo	Guamote	PALMIRA	No major complications with the collection.
Amazonia	Pastaza	Arajuno	ARAJUNO	No major complications with the collection.
Amazonia	Orellana	Loreto	LORETO	Authorities asked for a permit from the provincial delegation, delaying the data collection and generations distrust in some households. For the second phase, the team went to the outskirts of the community, due to the blockade from the authorities in the most populated part of the community.
Amazonia	Orellana	Loreto	SAN JOSE DE DAHUANO	Political authorities blocked the data collection process during the first phase. Lack of cellular coverage which limited the remote collection.
Amazonia	Orellana	Loreto	SAN JOSÉ DE PAYAMINO	Community eliminated for the second phase of listing due to lack of cellular coverage which limited the remote collection.
Amazonia	Orellana	Loreto	SAN VICENTE HUATICOCHA	Community eliminated for the second phase of listing due to lack of cellular coverage which limited the remote collection.
Amazonia	Orellana	Loreto	PUERTO MURIALDO	No major complications with the collection.

In line with the final databases used to develop this report, we specify any clarification in order to understand the obtained results:

FACE TO FACE

TOTAL REGISTERS	841
EFFECTIVE REGISTERS	762
NOT EFFECTIVE REGISTERS	79
DUPLICATED REGISTERS	47
DUPLICATED HHIDs	22
Effective registers with updated HHID (*HHID_CORREGIDO)*	2
Effective registers with the $4 mobile data	54%

**indicators (columns) added to the database:

- *HHID_CORREGIDO*: is the updated ID for each household, in case an enumerator added the wrong HHID.
- *Estado*: Based on our internal control and making sure every duplicated or inconsistent register is corrected, we specify if each register is effective or not effective.
- *Observaciones*: for those that apply, we detail a brief explanation of why the register is effective or not.

REMOTE:

TOTAL REGISTERS	2.395
EFFECTIVE REGISTERS	658
NOT EFFECTIVE REGISTERS	1.737
Registers in wrong community	47
Effective registers with the $4 mobile data	51%

**indicators (columns) added to the database:

- *HouseholdID union*: the original household ID (ID from data downloaded)
- *Codigo parroquia*: code created for each community using the data downloaded from the platform (Village or Camp level (Admin 5 Level))
- *Codigo parroquia con data original*: community code using the original databases sent with the listing of homes
- *Comparacion codigo parroquia*: we created these columns to match each household with its real community. If TRUE: original community = community from downloaded data
- *CODIGO generado*: HHID created with the household ID and the community code
- *CODIGO UNICO*: a unique code, not taking into account duplicated registers.
- *Estado*: Based on our internal control and making sure every duplicated or inconsistent register is corrected, we specify if each register is effective or not effective.
- *Observaciones*: for those that apply, we detail a brief explanation of why the register is effective or not.

MAIN CONCLUSIONS

As an overview of the study, we detail some conclusions regarding the remote and face-to-face data collection.

As the main observation, during the whole study, the effectiveness of the on-site survey was higher than for the remote survey. Even though at first sight, a call center survey may be simpler, the high degree of rurality of the target villages complicated the survey due to the low cell phone coverage. On the other hand, visiting households in person facilitated the availability and openness of the households surveyed in order to obtain a greater number of effective records.

It is also important to mention the limitations of rural communities in Ecuador, especially in the Sierra, where it was possible to perceive: a high level of distrust on the part of the people during the listing of homes and the study, caused by:

- Insecurity issues
- Distrust of people outside their community
- Blockages by local authorities to carry out the survey due to questioning about a possible collection of "sensitive" information of their village.

Due to this last limitation, we faced greater complications in the communities of the Sierra, where, on several occasions, we were unable to conclude with the sample required due to blockades by authorities that circulate immediately to their community.

Another factor to take into account when conducting a study in these types of communities is the lack of accessibility or availability of households that the enumerators face when conducting the study, both in person and remote. As a general complication, it was difficult to reach the respondent because he/she was at work or away from home. This increased the number of cases in which it was not possible to interview the respondent, but rather another member of the household. It is important to mention that in most cases, the most appropriate person to answer the interview, who is the head of household, is unavailable.

It is important to mention that, as a general comment we received from the enumerators, these is a strong openness and willingness from households since they understand that the study has a social assistance component and an organization dedicated to that. The credentials and introduction at the time of the interviews are fundamental for the households to understand the objective of the study.

A key step in this study is the listing of homes to be interviewed prior to the collection. Aligned with the level of rurality of the target communities, it is important to take into account that, when conducting a study and limiting it to households that have cell phones to be contacted for the remote survey, the target population is reduced to a group that has these facilities and eliminates a considerable number of households that could not participate in this crucial step.

As a closing remark, we summarize the limitations and advantages of each type of survey:

FACE-TO-FACE COLLECTION

General limitations

- Accessibility to the communities: remoteness of the villages, lack of public transportation, therefore, the logistics involved in carrying out the interview in person
- Insecurity issues
- Blockages from authorities
- Distrust of outsiders
- Dispersion of households: each community has its most concentrated area and the periphery.

General advantages

- Greater guarantee of an effective response
- Greater proximity at the time of the interview
- Easier to understand and communicate with respondent
- Possibility of making longer interviews

REMOTE COLLECTION

General limitations

- Lack of cell phone coverage
- Lack of response, more complex to get an interview
- High population without cell phones
- Low openness of people to talk on the phone
- Greater difficulty in understanding the questions and answers compared to face-to-face surveys
- A longer questionnaire may increase rejection from households

General advantages

- Simpler logistical aspects of the study
- Less possibility of facing blockades by authorities
- No risk of factors such as insecurity